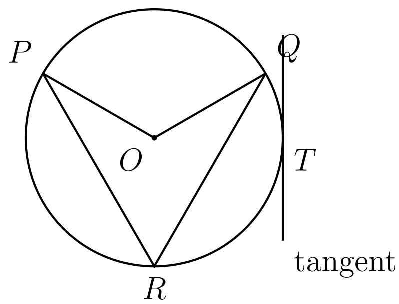
Putting the Circle Theorems Together

Central and inscribed angles, chord and tangent lengths, arc length and sector area, and circle equations

- After the warm-up the methods are mixed on purpose — decide which theorem the configuration calls for before you compute.
- Where a problem asks you to round, round only at the very end.



How every circle in this set is labelled: O is the centre, $\angle POQ$ is a central angle, $\angle PRQ$ is an inscribed angle on the same arc, and a tangent touches at one point such as T. No values shown — use the numbers given in each problem.

1. An inscribed angle of a circle intercepts an arc measuring 86°. Find the measure of the inscribed angle.

Answer: ________________

2. Inscribed angle $\angle ABC$ measures 37° and intercepts arc AC. Find the measure of arc AC.

Answer: ________________

3. A sector of a circle of radius 9 cm has a central angle of $40°$. Find the length of the sector's arc, to the nearest hundredth.

Answer: ________________ cm

4. Chords $\overline{AB}$ and $\overline{CD}$ of a circle intersect inside the circle at E. Given $AE = 6$, $EB = 10$ and $CE = 4$, find ED.

Answer: ________________

5. A circular fountain on a park map (axes in metres) has the equation $(x-5)^2+(y+2)^2 = 49$. A straight path runs tangent to the fountain at the point of the fountain's edge that lies directly above the centre.

(a) Find the radius of the fountain.

Answer: ________________ m

(b) Find the y-coordinate of the point where the path touches the fountain. ________

6. Quadrilateral $ABCD$ is inscribed in a circle, with $m\angle A = (2x + 15)^\circ$ and $m\angle C = (3x + 40)^\circ$.

(a) Find x. _________

(b) Find $m\angle A$. _________

7. A sector of a circle of radius 12 cm has a central angle of 120°. Find the area of the sector, to the nearest hundredth.

Answer: __________________ cm^2

8. From an external point P, a tangent segment touches a circle at T with $PT = 12$, and a secant from P cuts the circle at A then B, with the external part $PA = 8$. Find the whole secant length PB.

Answer: __________________

9. In a circle of radius 10 cm, minor arc AB measures $72°$.

(a) Find the length of minor arc AB, to the nearest hundredth.

Answer: ________________ cm

(b) Find the area of sector AOB, to the nearest hundredth.

Answer: ________________ cm^2

10. A circle has equation $x^2+y^2-10x+4y-71=0$, with x and y in units on a coordinate grid.

(a) Rewrite the equation in centre-radius form and give the radius. ________

(b) A chord of this circle is 16 units long. Find the distance from the centre to that chord.

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(c) Explain why this circle cannot contain a chord 24 units long.